

Alireza Kamali Ardakani

✉ alirezakamaliardakani@gmail.com

in [Alireza Kamali](#)

🔗 [alirezakamali80](#)

Education

BS University of Tehran, Mechanical Engineering

Sep 2020 - Sep 2025

- GPA: 3.83/4 (18.04/20) - Last two years: 3.92/4

- Ranked 9th/120

- Thesis: *Dynamic Analysis of Gough-Stewart Parallel Manipulators in a Serial Configuration*

SAMPAD High Schools, Mathematics

Aug 2017 - Aug 2020

- GPA: 4.0/4.0 (19.64/20)

- Ranked 1st/30

Research Interests

- Robotic Manipulation
- Autonomous Systems
- Biomechanics & Rehabilitation
- Dynamics and Control Systems
- Legged Locomotion
- Haptic Devices
- Mechatronics & Sensors
- Physics-Informed AI
- Bio-Inspired Mechanisms

Publications

Published and Accepted

- A. Kamali, H. Akbari, P. Namazian, M. T. Masouleh, A. Bahrami, "Dynamic Modeling of Double Segment Redundant Gough-Stewart Hybrid Manipulator Based on the Principle of Virtual Work," *2024 IEEE International Conference on Robotics and Mechatronics (ICROM)*.
- S. Rahmati, A. Kamali, P. Sobhani, A. Bahrami, "Design, Fabrication, and Control of a Mechanical Integrator," *Proceedings of the 2025 Iranian Society of Mechanical Engineering (ISME)* (accepted).

Manuscripts in Preparation

- A. Kamali, H. Akbari, M. T. Masouleh, "Reduced Dynamic Model of single Gough-Stewart Parallel Manipulator" (in preparation for 2025 IEEE International Conference on Robotics and Mechatronics (ICROM)).
- H. Akbari, A. Kamali, M. T. Masouleh, "Dynamic Modeling, Base Inertial Parameters, and Dynamic Sensitivity of Gough-Stewart Hyper-Redundant Manipulator" (in preparation for *Multibody System Dynamics*).

Experiences

Research Assistant

- Human-Robot Interaction Laboratory

University of Tehran - Iran

Apr 2024 - Present

Supervisor: Dr. Mehdi Tale Masouleh

- **Topic:** Dynamic modeling of Serial-Parallel Manipulators.
- Developed expertise in: Redundancy Resolution, Minimal Set Parameters
- Software: Matlab Simscape, Python, ADAMS

Teaching Assistant

Semester

Fall 2025
Fall 2024
Fall 2024
Spring 2024
Fall 2023
Spring 2023
Fall 2022

Course

Physics2
Robotics (Graduate Course)
Product Design and Development
Measurement Systems
Mechanical Vibrations
Numerical Methods
Mechanics of Materials 1

Instructor

Dr. Tale Masouleh
Dr. Hamed
Dr. A. Shirazi
Dr. Bahrami

Dr. Mahnama

University of Tehran – Iran

Projects

These and other projects are described in detail on my [GitHub account](#) .

Robotics Mini Projects - Robotics Course:

2024

Completed various robotics projects, including motion sensing, Scara robot analysis, ROS-based tasks, and deep learning for object detection.

Solenoid-Driven Rotary Motor with Closed-Loop Speed Control - Mechatronics Course:

2023

Designed and built a solenoid-driven rotary motor with closed-loop speed control, incorporating dynamic simulation, PI control, and real-time monitoring.

Mechanical Integrator:

2024

Designed, fabricated, and controlled a mechanical integrator using stepper motor-driven mechanisms, where input movement adjusts the lower shaft, transferring motion via a lead screw and linear bearing, with the final shaft's revolutions recorded as the integral's response.

Skills

Programming Languages: Python, C/C++

Software: MATLAB, Simulink - Simscape, SolidWorks, AutoCAD, ROS, Linux

Simulation: MSC Adams, Comsol, Proteus

Miscellaneous: Arduino, LaTeX

Artificial Intelligence: Machine Learning, Neural Networks, Deep Learning

Selected Courses

Courses	Scores (out of 4)	Courses	Scores (out of 4)
Robotics	4	Mechatronics	4
Automatic Control	4	Dynamics	4
Programming	4	Fundamentals of Electronics	4
Measurement Systems	4	Engineering Mathematics	4
MEMS/NEMS Introduction	4	Biomechanics	4

Language and Test Scores

- **Persian:** Native
- **English:** Professional Working Proficiency

Exam	Score	Test Date
IELTS	6.5	Nov 14 th , 2024

Honors and Awards

- Among the Top 15 winners of the **2024** and **2023** National Mechanical Engineering Student Olympiad.
- Ranked among the top 0.2% in the **2020** national university entrance exam.

Voluntary

Conference Organizer (Des 2023) : ICROM 2023 (International Conference of Robotics and Mechatronics)

Editor and Author (Sep 2020 - Sep 2021): Gaj International Publications – Chemistry

Teaching (Sep 2020 - Mar 2022) : self-employed Chemistry Teacher